CRYSTAL COOPER

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EDUCATION

Northwestern University **Expected Graduation: June 2027**

Bachelor of Science in Mechanical Engineering, Design Concentration, Entrepreneurship Minor GPA: 3.34/4.0

- Relevant Coursework: Machine Dynamics, Mechanical Design & Manufacturing, Fluid Mechanics, Thermodynamics, Mechanics of Materials, Electronics Design, Embedded Programming in Python, Material Science, Design Thinking and Communication 1 & 2, Industrial Design Projects 1, User Experience Design, Computer Integrated Manufacturing

MAJOR ENGINEERING PROJECTS

GEARBOX April 2026 – June 2026

- Led shaft design for two-stage gear reduction system transmitting 70 kW; evaluated bearing options against 5+ year service life requirement, determined initial selection insufficient, and switched to SKF 6313 deep-groove bearings with one planned replacement.
- Performed stress analysis and tolerance stack-up verification, achieving safety factor of 2.68 and validating infinite fatigue life

3D PRINTED BRAKE CALIPERS April 2025 – June 2025

- Translated safety standards (ISO 4210, EN 14781, CPSC) into design parameters; then ran two rounds of topology optimization and iterative FEA to minimize mass under a 2.75 mm deflection constraint.
- Built and tested two physical prototypes, identifying deflection as the critical constraint; increased cross-sectional stiffness to reduce deflection 7.5% while improving braking performance.
- Validated final design against all performance and compliance targets, achieving 12.2 m stopping distance with acceptable deflection under load

RIDE DESIGN PROJECT - EMV PROTOTYPE (TPED) April 2025 – June 2025

- Led track section design and fabrication for villain-themed EMV prototype
- Engineered curvature transitions to balance structural safety with immersive ride dynamics

SMARTREACH (DTC 2) March 2024 – June 2024

- Led user research and iterative design with wheelchair user in a 9-week testing cycle, discovering that accordion piece weight was the critical constraint
- Designed extendable pointer by hollowing accordion folds to reduce weight while maintaining structural integrity; validated final prototype enables independent ADA button access without wheelchair repositioning

LEADERSHIP & ORGANIZATIONS

THEME PARK ENGINEERING AND DESIGN GROUP (TPED) | Evanston, IL March 2024 – Present

CREATIVE DIRECTOR

- Led creative ideation and decision-making on projects like the Haunted House and Escape Room, fostering a team culture that emphasizes collaborative exploration and execution accountability

KAPPA ALPHA THETA | Evanston, IL January 2024 – Present

ONLINE MEDIA DIRECTOR

- Developed visual brand strategy and content system for Kappa Alpha Theta (120+ members)
- Established warm aesthetic for social media and community-centered narrative across platforms; developed content calendar and casting strategy prioritizing inclusive representation to encourage member participation

WORK EXPERIENCE

CROWN HOLDINGS | Tinley Park, IL June 2026 – August 2026

SUSTAINABILITY INTERN

- Designed GRI data management macro system with automated changelog, undo functionality, and guidance documentation; enabled non-technical team members to maintain sustainability reporting without errors
- Developed compliance frameworks across three systems: WASH facility scorecard (automated scoring), VOC standardization template (unified regional reporting), and GRI-compliant wastewater standards survey; improved data consistency across 100+ global facilities
- Created financial modeling framework establishing internal water pricing ($/cubic meter) and carbon cost valuation; evaluated ROI across 200+ CAPEX projects and identified high-impact initiatives for regional replication and CDP reporting
- Standardized 20by30 goals language and metrics across sustainability report, website, and brochure; developed data collection methodology with facility-level surveys to enable consistent global compliance tracking

SKILLS & INTERESTS

Technical Skills: Proficient in Excel, Python, MatLab, NX CAD, Onshape, Vizcom
Design Tools & Methods: Figma, Vizcom, Canva, user research, prototyping, design thinking, iterative testing
Languages: English (Native Language), Portuguese (Bilingual Proficiency), Spanish (Working Proficiency)
Shop Skills: 24" Di Arco bench Shear, Box and Pan Break, Delta 18" Floor Standing Drill Press, Vertical Band Saw, Laser Cutter